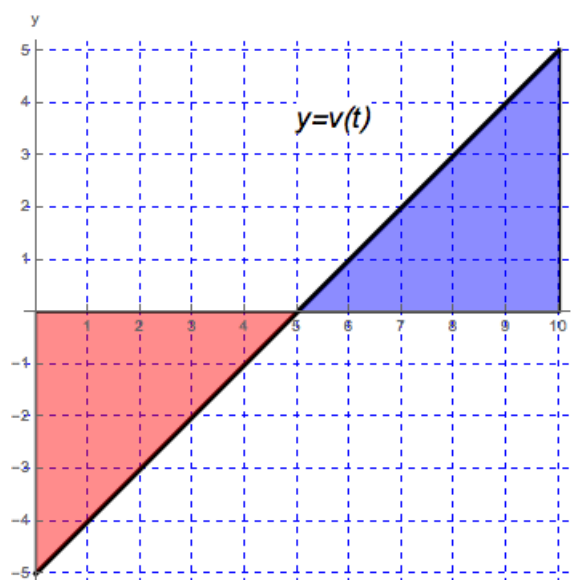


Problem 3

Problem 1 The velocity of the object moving along a straight line is given by $v(t) = t - 5$, $0 \leq t \leq 10$, where t is in seconds and v in ft/s.

- (a) Use geometry to evaluate the displacement of the object on the time interval $[0, 10]$. Sketch the graph of the velocity and shade the relevant region. [3]

Explanation. $s(10) - s(0) = \int_0^{10} (t - 5) dt$ = area under the curve and the interval $[0, 10]$ on the t -axis. Therefore $s(10) - s(0) = 0$ ft.

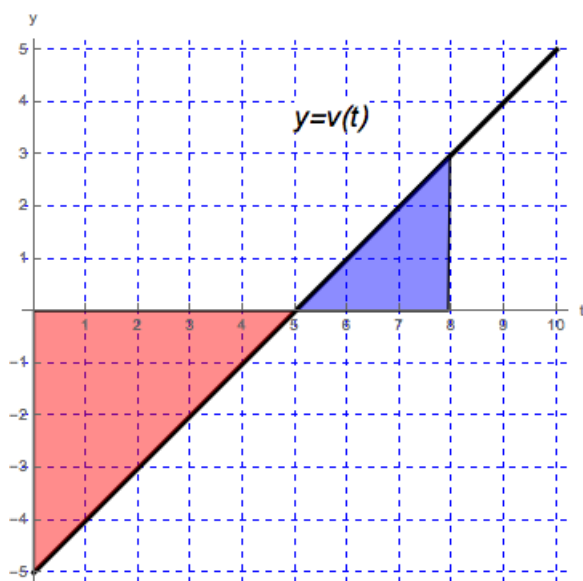


- (b) Compute the displacement on the interval $[0, 10]$ using antiderivatives of the function v .

Explanation. Since $s(t) = \frac{t^2}{2} - 5t + s(0)$, it follows that $s(10) - s(0) = \frac{10^2}{2} - 5(10) + s(0) - s(0) = 50 - 50 = 0$ ft.

- (c) Use geometry to evaluate the displacement of the object on the time interval $[0, 8]$. Sketch the graph of the velocity and shade the relevant region. [3]

Problem 3



Explanation.

$$s(8) - s(0) = \int_0^8 (t - 5) dt = \text{area under the curve and the interval } [0, 8]$$

on the t -axis. Therefore $s(8) - s(0) = -\frac{25}{2} + \frac{9}{2} = -8$ ft.

- (d) Compute the displacement on the interval $[0, 8]$ using antiderivatives of the function v .

Explanation. Since $s(t) = \frac{t^2}{2} - 5t + s(0)$, it follows that

$$s(8) - s(0) = \frac{8^2}{2} - 5(8) + s(0) - s(0) = 32 - 40 = -8 \text{ ft.}$$